Centripetal Force Examples

SYNOPSIS

An in-class sheet, not a lesson. One situation — a child on a merry-go-round — worked all the way from the spin rate to the force, and then repeated at a different radius so that you can see which quantities changed and which did not.

Everything you need is in Notes 1 and 2: ω from the rate of turn, $v_t = r\omega$, $a_c = v_t^2/r = r\omega^2$, and $F_c = ma_c$. The work is in getting the units right at the first step and then not losing track of which radius you are at.

A correction to the printed sheet. The stem on the paper copy reads "spinning at 10 rpm (10 rev/s)". **The parenthesis is wrong.** 10 rpm is 10 revolutions per *minute*, which is 0.167 rev/s — a factor of 60 away. Use 10 rev/min. Every number below follows from that, and if yours are 60 or 3600 times too big this is why.

BEFORE YOU START — PREDICT

Two children of the same mass ride the same merry-go-round, one at 2 m from the centre and one at 3 m. Write one sentence saying which of them has to hold on harder, and roughly by how much.

PERFORMANCE EXPECTATIONS

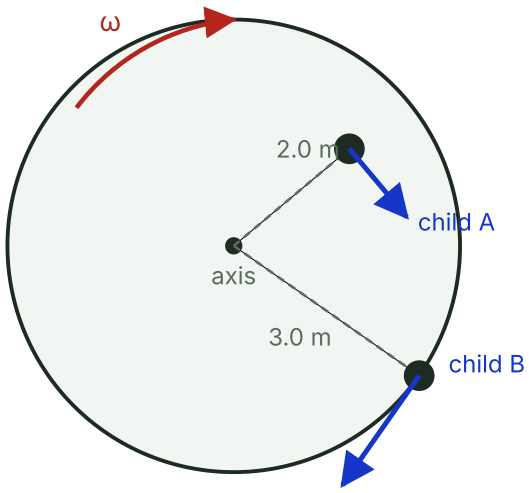
Convert a rate of turn in rev/min to an angular velocity in rad/s. Draw a top-down free-body diagram for a body in uniform circular motion and identify which real force is playing the centripetal role. Compute v_t, a_c and F_c at a given radius, and say how each one scales when the radius changes at fixed ω.

WATCH · THE SHEET WORKED THROUGH

Recording to come. Print the sheet (link at the bottom of this page), make your prediction, then work the items as the video goes through them.

The merry-go-round

A merry-go-round of radius 3.0 m turns at a steady **10 rev/min**. It is one rigid platform, so every point on it shares the same ω — and every point at a different radius has a different speed.



top-down view

one rigid platform

Both children share the same ω. Neither shares the same v_t, because $v_t = r\omega$ and their r differ.

Both velocity arrows are at right angles to their own radius, and the outer one is longer.

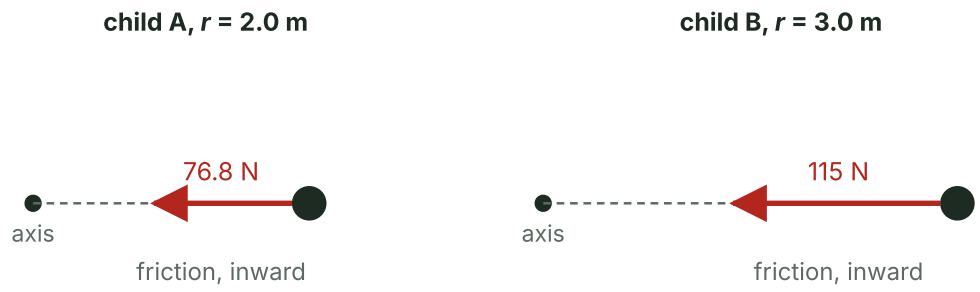
The platform top-down. The two radii are to scale: child B is at 1.5 times child A's radius, and her velocity arrow is 1.5 times as long.

1. Convert the rate of turn to an angular velocity. What is ω, in rad/s? (Watch the correction above — it is 10 revolutions per *minute*.)
2. Child A has mass 35 kg and sits 2.0 m from the centre. Find (A) her tangential speed v_t, in m/s; (B) her centripetal acceleration a_c, in m/s²; and (C) the centripetal force on her, in N.
3. Which *real* force on child A is the centripetal force? Say what happens to her if the merry-go-round speeds up past the point where that force can supply what is needed.

Show the working

The same child further out

Now move her, and change nothing else. The platform is rigid, so ω is the same for both positions — which is the whole reason this comparison is clean.



Top-down, so the weight and the normal force — both vertical — do not appear. Friction is the only horizontal force there is, so friction is the whole centripetal force. The arrows are drawn to scale.

The two free-body diagrams, top-down. Same child, same ω, same force *direction* — only the radius is different, and the arrow is exactly 1.5 times as long.

4. A second 35 kg child sits at the 3.0 m rim. Is holding on **easier, the same, or harder** for her than for child A? Work out v_t, a_c and F_c at 3.0 m and say by what factor each one changed.
5. Child A's seat is 2.0 m out and child B's is 3.0 m out. Without computing anything, say how v_t, a_c and F_c each depend on r at fixed ω — and check your answer against the numbers you just got.

Show the working

Looking ahead

Everything on this sheet was uniform circular motion: ω constant, so a_c constant and F_c constant. **Note 2** works the same method on situations where the centripetal force comes from something other than friction — a string, a track, gravity — and **Note 2b** applies it to the Moon.

BACK TO YOUR PREDICTION

The child **further out** has to hold on harder, and by exactly the ratio of the radii — 3.0/2.0 = 1.5, so 115 N against 76.8 N. If you guessed “harder” but expected a bigger factor, you may have been thinking of $a_c = v^2/r$ with v fixed; here it is ω that is fixed, because it is one rigid platform, and then the force is simply proportional to r.

Check yourself

1. A platform turns at 10 rev/min. What is ω, in rad/s?

_____ rad/s

2. Of the two children on this one rigid merry-go-round, the one further from the axis has

- ○ a larger ω and the same v_t
- ○ the same ω and a larger v_t
- ○ the same ω and the same v_t
- ○ a smaller ω and a larger v_t

3. What supplies the centripetal force on a child sitting on the floor of the merry-go-round?

- ○ her weight
- ○ the normal force from the floor
- ○ friction between her and the floor
- ○ a centrifugal force pushing her outward

4. What centripetal force acts on a 35 kg child 3.0 m from the axis at ω = 1.047 rad/s, in N?

_____ N

Check Clear

Your notes

IN CLASS

The sheet — blank (PDF)

print or open on a tablet; make your prediction, then work the items.

AFTER THE MODULE

Completed sheet (PDF)

released when Module 5 closes.

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